

Continuous Random Variables

Lecture 6

Topics in this lecture

- Probability Density Function (PDF)
- Cumulative Distribution Function (CDF)
- Normal Distribution

Continuous RV

- A random variable X is called continuous if there is a non-negative function $f_X(x)$, called probability density function (PDF) of X , such that

$$P(X \in B) = \int_B f_X(x) dx$$

for every subset B of the real line

- The probability $P(a \leq X \leq b)$ can be computed from the PDF as follows

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

$$P(X = a) = ?$$

- The probability $P(X = a)$ can be computed from the PDF as follows

$$P(X = a) = \int_a^a f_X(x) dx = 0$$

- Consequently,

$$P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$$

Properties of PDF

- All $f_X(x)$ must be non-negative

$$f_X(x) \geq 0 \quad , \quad \forall x$$

- Sum over all possibilities must be 1

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

Some Continuous RV

- Uniform random variable

$$f_X(x) = \begin{cases} \frac{1}{b-a} & , x \in [a, b] \\ 0 & , \text{otherwise} \end{cases}$$

- Exponential random variable

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & , \text{otherwise} \end{cases}$$

Example 6.1

- Let X be an exponential random variable with parameter $\lambda = 2$. Find

- $P(1 \leq X \leq 3)$

$$f_X(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$P(1 \leq X \leq 3) = \int_1^3 2e^{-2x} dx$$

$$= -e^{-2x} \Big|_1^3$$

$$= e^{-2} - e^{-6}$$

$$= 0.1329$$

Example 6.1

- Let X be an exponential random variable with parameter $\lambda = 2$. Find
 - $P(X > 5)$

$$\begin{aligned} & \int_5^{\infty} 2e^{-2x} dx \\ &= -e^{-2x} \Big|_5^{\infty} \\ &= e^{-10} \end{aligned}$$

Expected Value

- The expected value of a continuous random variable X can be obtained from its PDF as follows

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Example 6.2

- Let X be a uniform random variable in the interval $[1,5]$. Find $E[X]$

Expected Value of $g(X)$

- The expected value of $g(X)$ can be obtained from the PDF of X as follows

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Example 6.3

- Let X be a uniform random variable in the interval $[1,5]$. Let $Y = 2X + 3$. Find $E[Y]$

Computing Variance

- Variance of X can be computed as follows

$$\sigma_X^2 = E[X^2] - (E[X])^2$$

where

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Cumulative Distribution Function (CDF)

- A cumulative distribution function (CDF) of X is a function which describes the probability of the form $P(X \leq x)$
- If X is a continuous random variable, the CDF of X can be written as

con

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

- If X is a discrete random variable, the CDF of X can be written as

discrete

$$F_X(x) = \sum_{k=-\infty}^x p_X(k)$$

Example 6.4

- Let X be an exponential random variable with parameter λ . Find its CDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

$x \geq 0$

$$F_X(x) = \int_0^{\infty} \lambda e^{-\lambda t} dt$$

$$\begin{aligned} F_X(x) &= -e^{-\lambda t} \Big|_0^x \\ &= 1 - e^{-\lambda x} \end{aligned}$$

Properties of CDF

1. $F_X(x)$ is a monotonically non-decreasing function $\rightarrow$ If $x \leq y$, then $F_X(x) \leq F_X(y)$
2. $F_X(x)$ approaches 0 as x approaches $-\infty$
3. $F_X(x)$ approaches 1 as x approaches ∞
4. If X is discrete, then is a piecewise constant function of x
5. If X is continuous, then is a continuous function of x
6. If X is discrete, then

$$p_X(k) = P(X \leq k) - P(X \leq k - 1) = F_X(k) - F_X(k - 1)$$

7. If X is continuous, then for those x at which the PDF is continuous

$$f_X(x) = \frac{dF_X}{dx}(x)$$

Normal Random Variable

- A continuous random variable X is normally distributed if it has the PDF of the following form

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, \quad -\infty < x < \infty$$

where μ is the mean and σ^2 is the variance of X

- Standard normal, when $\mu = 0$ and $\sigma^2 = 1$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty$$

Transforming Normal to Standard Normal

- If X is normally distributed with mean μ and variance σ^2 , we can create a new standard normal random variable Z from X as follows

$$Z = \frac{X - \mu}{\sigma}$$

- Z is a linear transformation of a normal random variable X , so Z is also normal. The mean and variance of Z are as follow.

$$E[Z] = \frac{E[X] - \mu}{\sigma} = 0$$

$$VAR[Z] = \frac{VAR[X]}{\sigma^2} = 1$$

Computing Probability from Standard Normal

- If Z is a standard normal variable, we can find $P(Z \leq z)$, defined as $\Phi(z)$, from a standard normal table

Standard Normal Probabilities

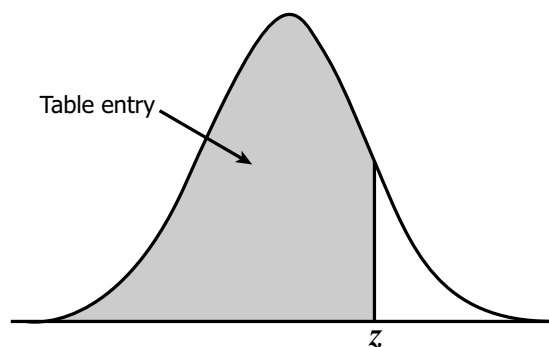


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545

Probability Relating to Normal RV in General

- If X is normally distributed with mean μ and variance σ^2 , we use these steps to compute the probability $P(X \leq x)$
 - Create new RV, $Z = (X - \mu)/\sigma$
 - Find the probability $P(X \leq x)$ from the standard normal table as follows

$$\begin{aligned} P(X \leq x) &= P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) \\ &= P\left(Z \leq \frac{x - \mu}{\sigma}\right) \\ &= \Phi\left(\frac{x - \mu}{\sigma}\right) \end{aligned}$$

Example 6.5

- If X is normally distributed with mean $\mu = 2$ and variance $\sigma^2 = 4$, find the probability $P(X \leq 3)$

Standard Normal Probabilities

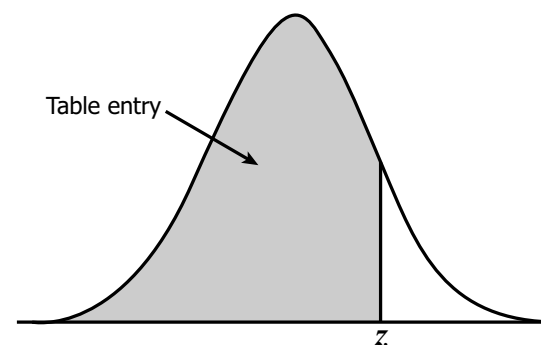


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0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
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Example 6.6

- If X is normally distributed with mean $\mu = 2$ and variance $\sigma^2 = 4$, find the probability $P(1.5 \leq X \leq 3)$

$$\frac{3-2}{2}$$

Answer

Standard Normal Probabilities

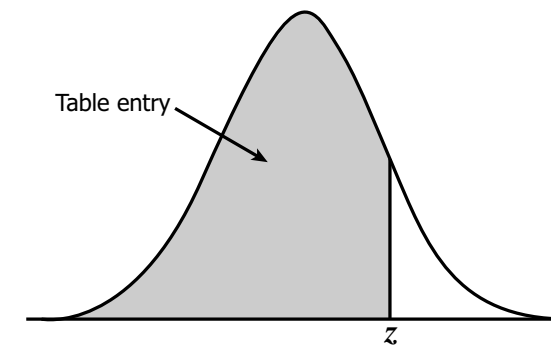


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